The Virtual Bingo Blower: An open-source tool to generate ambiguity and risk in experiments*

Ola Andersson† Geoffrey Castillo‡ Erik Wengström§

11th November 2026

This paper introduces the Virtual Bingo Blower (VBB), an open-source tool to generate risk and ambiguity in computerized experiments. The VBB simulates a conventional bingo blower in real time, and the number, colour, size, and speed of its balls can be easily modified. We validate the VBB in an online experiment that manipulates the number of balls while keeping the proportions of colours fixed. With 60 balls instead of 10, subjects' matching probabilities are much less accurate, showing that the VBB can vary how hard the proportions of colours are to judge.

Keywords: ambiguity, bingo blower, experimental tools, online experiment, randomization, risk
JEL Codes: C91, D81

*We gratefully acknowledge financial support from the Jan Wallander and Tom Hedelius Foundation and Riksbankens Jubileumsfond. We are grateful for comments from John Hey, Gianna Lotito, Anna Maffioletti, Andrea Morone, and Peter Wakker, as well as seminar participants at ESA World 2023 in Lyon, NBEE in Lund, MiddExLab virtual experimental economics seminar, Department of Psychology at Uppsala University, and ESA Europe 2024 in Helsinki.

†Department of Economics, Uppsala University, Box 513, 751 20 Uppsala, Sweden. ola.andersson@nek.uu.se

‡Corresponding author. BEAR and Department of Economics, Nottingham Trent University, 50 Shakespeare Street, Nottingham NG1 4FQ, United Kingdom. geoffrey.castillo@ntu.ac.uk

§Department of Economics, Lund University, P.O. Box 118, 221 00 Lund, Sweden. erik.wengstrom@nek.lu.se

1 Introduction

Ambiguity refers to situations in which the probabilities attached to the outcomes are unknown. We often make decisions under ambiguity, for example when looking for a job, choosing a medical treatment, or investing in the stock market. Due to its importance, many experiments have been conducted to test theories of choice under ambiguity or to measure attitudes towards ambiguity (Camerer and Weber, 1992; Trautmann and van de Kuilen, 2015). A key question when designing such experiments is how to generate ambiguity in a way that is both intuitive and credible.

We present a new method for generating ambiguity in experiments: the Virtual Bingo Blower (VBB). The VBB is inspired by physical bingo blowers, which have become popular for generating ambiguity in experiments (Georgalos, 2019, 2021; Hey et al., 2010; Hey and Pace, 2014; Kothiyal et al., 2014). In a bingo blower, balls of different colours move continuously, and a randomly selected ball determines the outcome. The constant motion of the balls makes it difficult to assess the proportions of colours, thereby generating ambiguity. Bingo blowers are an intuitive and transparent way of generating ambiguity, but they are hard to deploy at scale as they are bulky, noisy, and ill-suited for online experiments.

The VBB addresses the limitations of physical bingo blowers by being an open-source tool that runs directly in any modern web browser, including on mobile devices. It can thus be used in online and laboratory experiments with any web-based experimental software, such as Qualtrics, oTree (Chen et al., 2016), or LIONESS (Giamattei et al., 2020). The VBB is available as `bingo-blower.js` at `https://github.com/geoffreycastillo/bingo-blower-js`, and a demo is available at `https://geoffreycastillo.com/bingo-blower-js-demo`.

The main advantage of the VBB over other non-physical devices is the use of a real-time physics engine that simulates the motion of balls according to physical laws. The physics simulation runs locally on participants' computers, independent of any input from experimenters. As a result, the movement of the balls is unpredictable and cannot be manipulated or anticipated by anyone, ensuring that ambiguity remains transparent and credible. Anyone can check in the open-source code that no function can force or even influence the draw. In any case, the evidence indicates that the potential for manipulability, a common concern with virtual devices, does not influence elicited ambiguity attitudes (Li et al., 2024; Oechssler and Roomets, 2015).

The VBB is also flexible. The proportions of colours can be made harder or easier to judge by changing features such as the number, colour, or speed of the balls, even

in-between trials. With a number of balls that is easily countable, the VBB can also generate risk. It can further replace opaque random number generators, for example to play out the lottery selected for payment in a risk elicitation task, or to choose which round of a public goods game is paid.

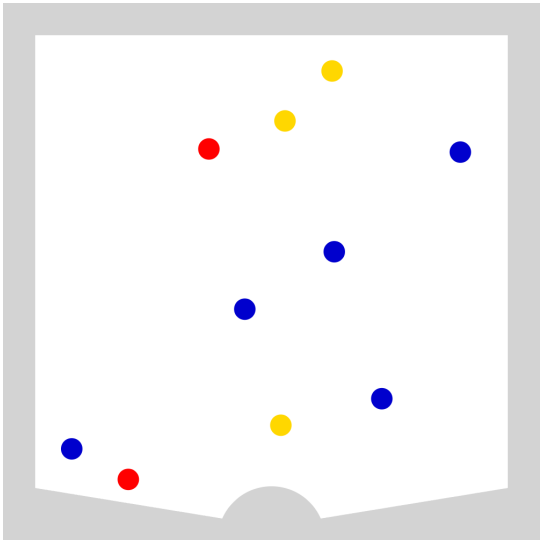
We validate the VBB in an online experiment in which we elicit matching probabilities (Baillon et al., 2018) with a blower of either 10 or 60 balls, keeping the proportions of colours fixed. With 60 balls, matching probabilities are much less accurate, and subjects hardly distinguish between the colours. Thus, increasing the number of balls is enough to make the composition of the VBB harder to judge. The ambiguity-generated likelihood insensitivity index (a-index) of Baillon et al. (2018) is also slightly higher with 60 balls, though not robustly so.

As Hey et al. (2010) explain, bingo blowers avoid the information asymmetry between experimenters and subjects of unknown urns (made famous by Ellsberg, 1961). Subjects can see the balls moving, and can witness how a ball is selected for payment. Other attempts have been made to computerize bingo blowers. Levati and Morone (2013) projected a 3D animation of a bingo blower cage on a wall, and Morone and Caferra (2024) developed the Ambiguity Box, which Caferra et al. (2023) used to study dynamic inconsistency under ambiguity. The Ambiguity Box is a sequence of pre-rendered static images displayed as a GIF, in which the proportions of squares of different colours correspond to the underlying probabilities. By contrast, the VBB simulates the movement of the balls in real time, which makes it closer to the experience of a real bingo blower. It also lets subjects trigger and witness the draw that determines their earnings, whereas the Ambiguity Box relies on a random number generator to play out the chosen lottery, which reduces transparency to subjects.

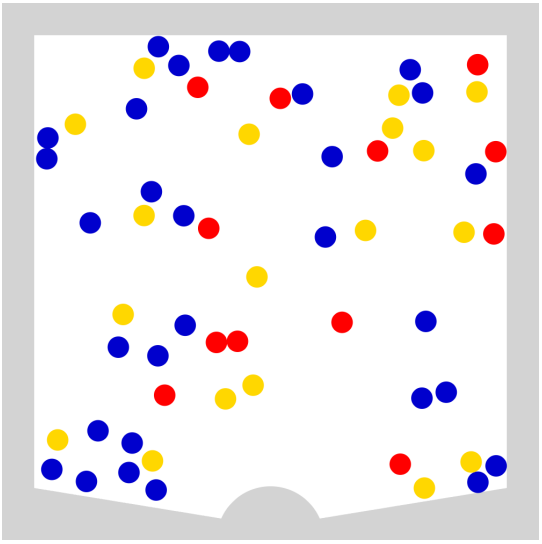
2 The Virtual Bingo Blower as an experimental tool

Adding the VBB to an experiment requires very little programming. In the simplest case, two lines of JavaScript are enough: `let blower= new BingoBlower();` creates a blower, and `blower.addBalls([2,5,3]);` fills it with 2, 5, and 3 balls of three different colours. The GitHub repository provides documentation and example code and describes all the options in detail.

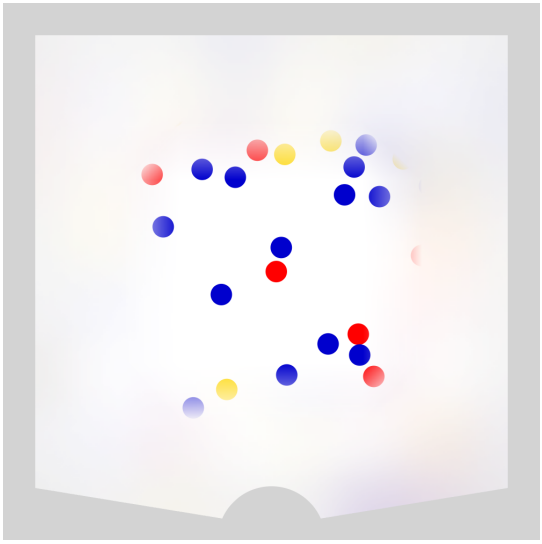
Researchers can change the blower in several ways, some of which are illustrated in Figure 1. First, they can change its composition: the number of balls, the number of colours, and their proportions. Second, they can make the balls easier or harder to see,



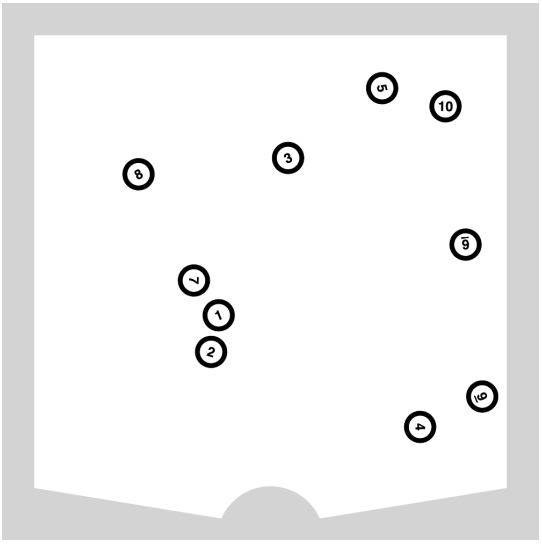
(a) 10 balls



(b) 60 balls



(c) 60 balls with frosted edges



(d) 10 numbered balls

Figure 1: Examples of VBB variations generated by changing software parameters. Panels (a) and (b) keep the proportions of colours fixed at 20% red, 50% blue, and 30% yellow. In panel (c), balls near the walls are blurred and washed out. Panel (d) shows 10 numbered balls.

by changing the size of the display, the size of the balls, or the colours. Third, they can change how the balls move: a stronger simulated air flow or bouncier balls make the motion faster and harder to follow. Finally, they can frost the edges of the blower so that balls near the walls cannot be counted, even from a screenshot (Figure 1c). All these changes make the proportions of colours harder or easier to judge while keeping the actual proportions fixed. In the experiment below we test the simplest of them, the number of balls.

The VBB also has a built-in procedure to draw a ball for payment. When participants click a button, a target appears in the blower and the balls keep moving for a few seconds. The simulation then stops, and the ball closest to the target is selected. Its colour, or its number if the balls are numbered (Figure 1d), determines the participant's payoff. Participants can also move the balls with the mouse or on a touchscreen, to check for themselves that the balls follow the physics simulation.

3 Experimental design

We test whether changing the number of balls in the VBB makes the proportions of colours harder to judge, and whether it changes ambiguity attitudes. In two between-subject treatments, the blower has either 10 balls (VBB10) or 60 balls (VBB60), with the proportions of colours fixed at 20% red, 50% blue, and 30% yellow.

We elicit matching probabilities with the method of Baillon et al. (2018), to which we refer for details. The set of events is partitioned into three single events corresponding to the colours in the VBB and their three pairwise unions (the composite events). For each of these six events, subjects report, through a choice list, the matching probability m that makes them indifferent between winning $x = £3$ if the event obtains and winning x with probability m; Figure A.4 in Online Appendix A shows an example.

We recruited participants on Prolific on the basis of a preregistered power calculation with effect sizes as in Baillon et al. (2018), targeting 150 per treatment and over-recruiting to allow for exclusions.[1] Participants had to live in the UK, be fluent in English, and have a Prolific approval rating of at least 95%; they received a £1 fee plus a £3 bonus if a winning event occurred. The experiment was programmed using oTree (Chen et al., 2016), and the full instructions and task screenshots are in Online Appendix A.[2]

[1]The experiment also included a natural-events treatment, which we analyze in a companion paper.

[2]Our oTree code is available at `https://github.com/geoffreycastillo/otree-virtual-bingo-blower`. Our design, hypotheses, and exclusion criteria were preregistered on the OSF: `https://osf.io/dm3kq`.

As preregistered, we exclude participants who used a mobile phone or whose frame rate was below 50 FPS.[3] As in Baillon et al. (2018), we also exclude participants with a missing matching probability: 58 chose a dominated option at the top or bottom of at least one choice list, and the remaining one switched more than once. We are left with 317 subjects across the two VBB treatments: 162 in VBB10 and 155 in VBB60.[4]

4 Results

Our first result is that subjects' matching probabilities are less accurate when the VBB has more balls. Because the VBB fixes the proportions of colours, we know the true probability of every event: 20%, 30%, and 50% for the single events, and 50%, 70%, and 80% for the composite events.

Figures 2 and 3 show the distributions of the matching probabilities. In VBB10, the matching probabilities of each event peak in the interval just above its true probability. In VBB60, the distributions of the three events overlap and peak close to 1/3 for the single events and 2/3 for the composite events. Figure 4 shows that mean matching probabilities increase steadily with the true probabilities in VBB10, but are much flatter in VBB60.

Three tests confirm that subjects in VBB60 are less accurate. First, the mean absolute error between a subject's matching probabilities and the true probabilities of the single events is 11.2 percentage points in VBB10 and 15.6 in VBB60 (Mann–Whitney test, two-sided $p < 0.001$).[5] Second, 50.6% of subjects in VBB10 rank the three single events in the right order, against 18.1% in VBB60, close to the 16.7% expected under random ranking (Fisher's exact test, two-sided $p < 0.001$). Third, 31.5% of subjects in VBB10 report matching probabilities equal to the true probabilities for all three single events, against 0.6% in VBB60 (Fisher's exact test, two-sided $p < 0.001$). This suggests that about a third of subjects in VBB10 counted the balls. Results for the composite events are similar and are reported in Online Appendix C.1.

Our second result is that the a-index of Baillon et al. (2018) is slightly higher with more balls. We compute both indices from the same matching probabilities: the b-index, which measures ambiguity aversion ($b = 0$ under ambiguity neutrality, $b > 0$ aversion, $b < 0$ seeking), and the a-index, which measures likelihood insensitivity ($a = 0$ under

[3] Online Appendix D explains this criterion. We control for the frame rate in model 3 of Online Appendix Table C.1.

[4] Online Appendix B reports the demographic characteristics of our sample.

[5] Matching probabilities are midpoints of switching intervals and every true probability is an interval edge, so the smallest possible mean absolute error is 2.5.

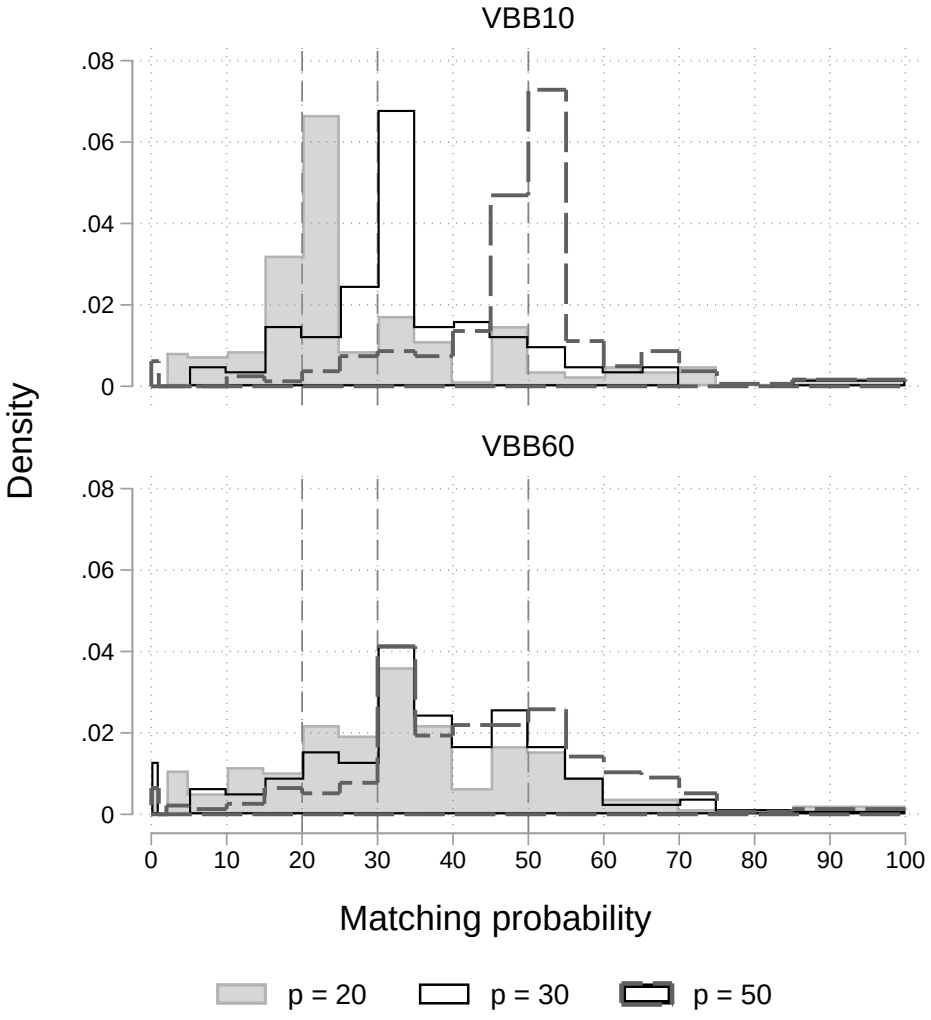


Figure 2: Distribution of matching probabilities for the single events, by treatment. Dashed lines mark the true probabilities (20%, 30%, and 50%).

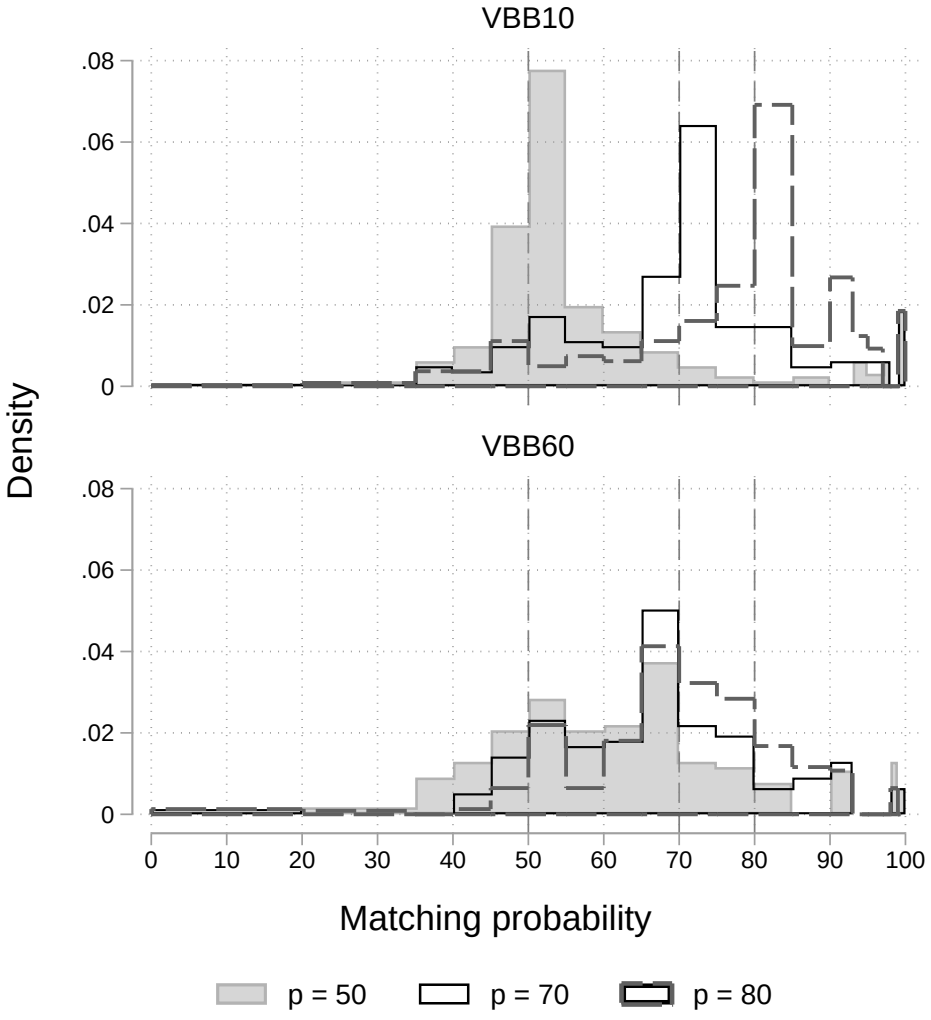


Figure 3: Distribution of matching probabilities for the composite events, by treatment. Dashed lines mark the true probabilities (50%, 70%, and 80%).

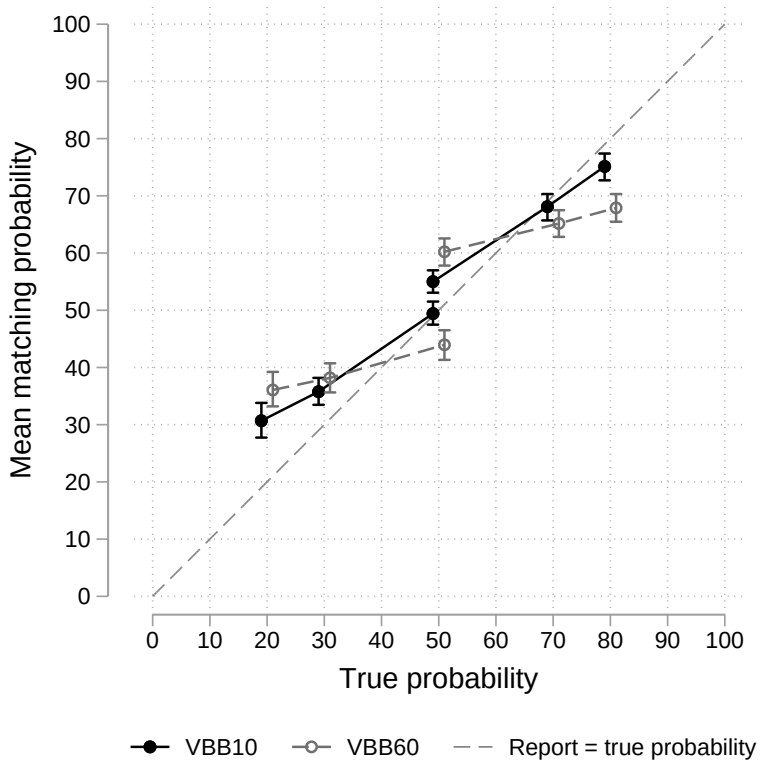


Figure 4: Mean matching probability of each event against its true probability, by treatment, with bootstrapped 95% confidence intervals (2,000 replications).

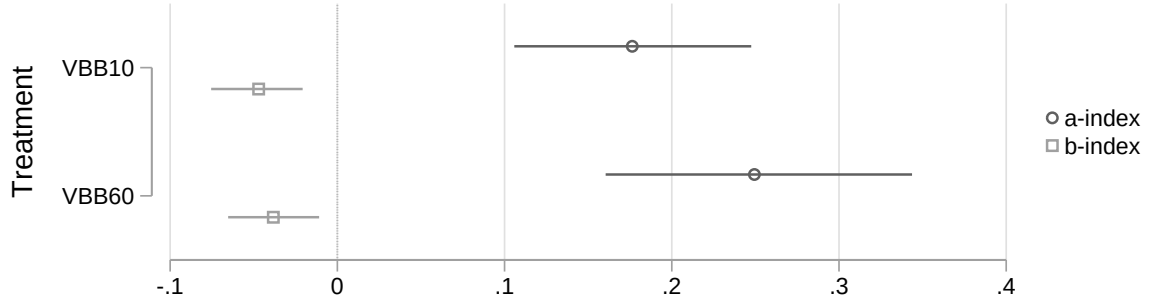


Figure 5: Mean a-index and b-index in each treatment, with bootstrapped 95% confidence intervals (2,000 replications, resampling within each treatment).

ambiguity neutrality). We preregistered no difference in the b-index between treatments, and a higher a-index in VBB60.

Figure 5 shows the mean of each index by treatment. We run the preregistered OLS regressions of each index on a treatment dummy, without controls (model 1) and with controls for age and sex (model 2), with robust standard errors (Online Appendix Table C.1). For the directional a-index hypothesis, we report both one-sided and two-sided p-values; all other p-values are two-sided.

The b-index does not differ between the two treatments ($p = 0.659$ without controls and $p = 0.736$ with controls). The a-index is 0.073 higher in VBB60 than in VBB10 without controls, and 0.094 higher with controls. Without controls, the difference is not significant (one-sided $p = 0.105$, two-sided $p = 0.210$). With controls, it is significant at the 10% level with a one-sided test ($p = 0.051$). Preregistered Mann-Whitney tests find no difference between the treatments for either index ($p = 0.765$ for the b-index; one-sided $p = 0.206$, two-sided $p = 0.413$ for the a-index).

That the a-index is only slightly higher in VBB60 is not surprising given our first result. The indices only use the average matching probabilities of the single and of the composite events, so they are zero whenever matching probabilities are additive, whatever beliefs they reflect (Baillon et al., 2018). For instance, reporting the true probabilities and reporting 1/3 for each single event and 2/3 for each composite event both give $a = b = 0$.[6] The second pattern is close to what we observe in VBB60 (Figures 2 to 4). Subjects in VBB60 hardly distinguish between the colours, but they still separate the single from the composite events about as well as subjects in VBB10 (Figure 4). More balls thus mainly blur subjects' beliefs about the composition of the VBB, which the a-index is designed to net out.

[6]With our choice lists, which record the midpoint of the switching interval, the first subject gets $a \in [-0.15, 0.15]$ and the second $a = -0.05$.

Finally, subjects' free-text comments show no sign of suspicion about the VBB, a possible worry for any virtual randomization device. We classified the comments into engagement, understanding, technical, and neutral categories (Online Appendix C.4). None expressed suspicion, and many indicate that subjects found the VBB engaging. Since we did not elicit trust directly, this shows an absence of spontaneous suspicion rather than a direct measure of trust. Online Appendix D discusses further ways to mitigate suspicion, alongside a note on device performance.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to write parts of the analysis code and to draft, shorten, and edit parts of the manuscript and its highlights. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

Baillon, Aurélien, Zhenxing Huang, Asli Selim, and Peter P. Wakker (2018) "Measuring ambiguity attitudes for all (natural) events," *Econometrica*, 86 (5), 1839–1858, 10.3982/ECTA14370.

Caferra, Rocco, John D. Hey, Andrea Morone, and Marco Santorsola (2023) "Dynamic inconsistency under ambiguity: An experiment," *Journal of Risk and Uncertainty*, 67 (3), 215–238, 10.1007/s11166-023-09418-y.

Camerer, Colin and Martin Weber (1992) "Recent developments in modeling preferences: Uncertainty and ambiguity," *Journal of Risk and Uncertainty*, 5 (4), 325–370, 10.1007/BF00122575.

Chen, Daniel L., Martin Schonger, and Chris Wickens (2016) "oTree—An open-source platform for laboratory, online, and field experiments," *Journal of Behavioral and Experimental Finance*, 9, 88–97, 10.1016/j.jbef.2015.12.001.

Ellsberg, Daniel (1961) "Risk, ambiguity, and the Savage axioms," *The Quarterly Journal of Economics*, 75 (4), 643–669, 10.2307/1884324.

Georgalos, Konstantinos (2019) "An experimental test of the predictive power of dynamic ambiguity models," *Journal of Risk and Uncertainty*, 59 (1), 51–83, 10.1007/s11166-019-09311-7.

——— (2021) "Dynamic decision making under ambiguity: An experimental investigation," *Games and Economic Behavior*, 127, 28–46, 10.1016/j.geb.2021.02.002.

Giamattei, Marcus, Kyanoush Seyed Yahosseini, Simon Gächter, and Lucas Molleman (2020) "LIONESS Lab: A free web-based platform for conducting interactive experiments online," *Journal of the Economic Science Association*, 6 (1), 95–111, 10.1007/s40881-020-00087-0.

Hey, John D., Gianna Lotito, and Anna Maffioletti (2010) "The descriptive and predictive adequacy of theories of decision making under uncertainty/ambiguity," *Journal of Risk and Uncertainty*, 41 (2), 81–111, 10.1007/s11166-010-9102-0.

Hey, John D. and Noemi Pace (2014) "The explanatory and predictive power of non two-stage-probability theories of decision making under ambiguity," *Journal of Risk and Uncertainty*, 49 (1), 1–29, 10.1007/s11166-014-9198-8.

Kothiyal, Amit, Vitalie Spinu, and Peter P. Wakker (2014) "An experimental test of prospect theory for predicting choice under ambiguity," *Journal of Risk and Uncertainty*, 48 (1), 1–17, 10.1007/s11166-014-9185-0.

Levati, M. Vittoria and Andrea Morone (2013) "Voluntary contributions with risky and uncertain marginal returns: The importance of the parameter values," *Journal of Public Economic Theory*, 15 (5), 736–744, 10.1111/jpet.12043.

Li, Jiangyan, Kim Fairley, and Achiel Fenneman (2024) "Does it matter how we produce ambiguity in experiments?" MPRA Paper 122336, `https://ideas.repec.org//p/pra/mprapa/122336.html`.

Morone, Andrea and Rocco Caferra (2024) "The Ambiguity Box: A new tool to generate ambiguity in the lab," *Journal of Behavioral and Experimental Economics*, 113, 102299, 10.1016/j.socec.2024.102299.

Oechssler, Jörg and Alex Roomets (2015) "A test of mechanical ambiguity," *Journal of Economic Behavior & Organization*, 119, 153–162, 10.1016/j.jebo.2015.08.004.

Trautmann, Stefan T. and Gijs van de Kuilen (2015) "Ambiguity attitudes," in *The Wiley Blackwell Handbook of Judgment and Decision Making*, Chap. 3, 89–116: John Wiley & Sons, Ltd, 10.1002/9781118468333.ch3.